

Shaik Ibrahim Kalilulla

Narasaraopet, Andhra Pradesh, India

Email: ibrahimkalilulla413@gmail.com

Phone: +91 8978929938

EXPERIENCE:

Electronics Hardware Lab Technician,
Narasaraopeta Institute of Technology

May2018 –April2022

- Install, calibrate, and maintain lab instruments like oscilloscopes, multi-meters, power supplies, signal generators, etc.
- Ensure tools and test equipment is in working order.
- Developed and implemented laboratory exercises using VHDL/Verilog, enhancing students' practical understanding of FPGA design concepts.
- Assisted in the synthesis and implementation of student projects using Vivado, ensuring successful deployment of FPGA designs.
- Conducted functional and timings simulations for academic projects, resulting in improved accuracy and reliability of student designs.
- Debugged and resolved design issues in student projects using Vivado Logic Analyzer, leading to optimized design performance and successful project outcomes.
- Documented lab procedures and project outcomes, creating comprehensive guides that improved student comprehension and streamlined future lab sessions.
- Designed and implemented IoT solutions using sensors and microcontrollers, enabling real-time data collection and communication across networks.
- Developed firmware for embedded systems in IoT devices, ensuring efficient operation and low-power consumption.
- Integrated IoT devices with cloud platforms for remote monitoring, control, and data analysis, improving the scalability and accessibility of projects.
- Collaborated with cross-functional teams to implement secure communication protocols in IoT applications

Electronics Hardware Lab Technician,
Narasaraopeta Engineering College

May2023 –May2025

- Install, calibrate, and maintain lab instruments like oscilloscopes, multi-meters, power supplies, signal generators, etc.
- Ensure tools and test equipment is in working order.
- Developed and implemented laboratory exercises using VHDL/Verilog, enhancing students' practical understanding of FPGA design concepts.
- Assisted in the synthesis and implementation of student projects using Vivado, ensuring successful deployment of FPGA designs.
- Conducted functional and timings simulations for academic projects, resulting in improved accuracy and reliability of student designs.
- Debugged and resolved design issues in student projects using Vivado Logic Analyzer, leading to optimized design performance and successful project outcomes.
- Documented lab procedures and project outcomes, creating comprehensive guides that improved student comprehension and streamlined future lab sessions.
- Designed and implemented IoT solutions using sensors and microcontrollers, enabling real-time data collection and communication across networks.
- Developed firmware for embedded systems in IoT devices, ensuring efficient operation

and low-power consumption.

- Integrated IoT devices with cloud platforms for remote monitoring, control, and data analysis, improving the scalability and accessibility of projects.
- Collaborated with cross-functional teams to implement secure communication protocols in IoT applications

Technical Skills & Tools & Softwares:

- Oscilloscopes, multimeters, logic analyzers
- Soldering stations, hot air guns.
- PCB design tools (e.g., KiCAD, Eagle, Altium – for basic edits or checks)
- Lab management tools (e.g., inventory tracking spreadsheets or software)
- VLSI Design & Embedded Systems: VLSI Design, Embedded Systems, Digital Circuit Design, PCB Designing
- Programming Languages & Tools: C- Language, HDL, Verilog, ARM Cortex
- Design & Simulation Tools: Xilinx Vivado, Mentor Graphics, Cadence, EDA Tools, MATLAB, Simulink, Multisim, LTspices, Arduino IDE
- Microcontroller & Embedded Protocols: Microcontroller, Keil, I2C, SPI, UART, USB, RS232.

EDUCATION:

Narasaraopeta Institute of Technology, Narasaraopet
M.Tech in VLSI & Embedded Systems

September 2018–June 2021

Qis College of Engineering and Technology, Ongole
B.Tech in Electronics and Communication Engineering

August 2014–May 2017

COURSE CERTIFICATIONS:

- | | |
|--|-----------------------------------|
| 1. 'C' Programming
Brilliant Computer Education, Ongole | December 2015–January 2016 |
| 2. Diploma in Computer Applications
Micro Line Computer, Guntur | May 2016 –October 2016 |
| 3. Programming for Everybody (Getting Started with Python)
Credential URL: Coursera Certificate | May 2020 |
| 4. Introduction to Artificial Intelligence (AI)
Credential URL: Coursera Certificate | May 2020 |
| 5. PCB Design Software Installation Essentials
Credential URL: https://www.mindluster.com/storage/cer/6e96763b.jpg | December 2024 |
| 6. Protocol SPI Fundamentals
Credential URL: https://www.mindluster.com/student/certificate/c411f63b | December 2024 |